DPP DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CENTRAL VENOUS ACCESS DEVICE MANAGEMENT
DEPARTMENT:	INTENSIVE CARE UNIT
APPLIES TO:	ALL MEDICAL AND NURSING STAFF

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1. PURPOSE

1.1. To provide procedural guidelines in assisting the physician in CVAD insertion, Care, Routine Flushing and Removal

1.2. To provide procedural guidance to all staff on care and use of CVAD including dealing with any problems encountered.

1.3. To prevent infection in collaboration with infection Control in patients with CVAD.

1.4. To establish protocol on how to maintain patency of Central Venous Access Device.

2. DEFINITION & ABBREVIATION

2.1. **Central Venous Access Device (CVAD)** - also known as Central Line, Central Venous Line or Central Venous Catheter. It is a multipurpose venous catheter that terminates at or close to the heart or in one of the great vessels. It is placed into a large vein in the neck (Internal Jugular Vein), chest (Subclavian Vein or Axillary Vein), and groin (Femoral Vein).

2.2. **Aseptic Technique** -a procedure used by medical staff to prevent the spread of infection

2.3. **PPE (Personal Protective Equipment)** - an equipment that will protect the staff against health or safety risks at work

2.4. **Hand Hygiene (HH)** - a general term that applies to routine hand washing, antiseptic hand wash, antiseptic hand rub, or surgical hand antisepsis.

2.5. **Valsalva maneuver** - forced expiratory effort against a closed glottis that decreases intrathoracic pressure, hampering venous return to the heart, and that can be used to inflate the Eustachian tubes and adjust pressure in the middle ear.

2.6. **MCC** – Medica Cloud Care System (HIS – Hospital Information System)

3. RESPONSIBILITY

3.1. Physician and Staff Nurses must be competent in managing CVAD according to the policy.

3.2. Physician and Staff Nurse utilize strict aseptic technique.

3.3. The ICU Head Nurse ensures that all staff is competent in assisting CVAD insertion, care, flushing and removal. Witnesses obtaining of CRITICAL CARE CONSENT FORM by Attending Physician/ Medical Specialist On Call/ Designee in the ward/unit.

4. POLICY

4.1. A complete physician's written order must be written.

4.2. Consent must be secured and informed prior to insertion except emergency situation

4.2.1 In case of emergency situation, physician who will insert the CVAD should sign the consent form and documentation in the progress notes must be done.

4.3. Patient must be on continuous cardiac monitoring.

4.4. Strict compliance with the Infection Control Guidelines when performing the procedure.

4.5. CVAD dressing must be changed only when the dressing becomes damp, loosened, or soiled. Do not change dressings at routinely scheduled intervals.

4.6. Two trials only allowed in one site for CVAD insertion.

4.7. CVADs are removed without delay once the physician decides they are no longer required for patient care.

5. PROCEDURE

5.1. **CVAD Insertion, Assisting**

5.1.1. Physician writes order for CVAD insertion.

5.1.2. Explain the procedure to the patient.

5.1.3. Gather all equipment needed and establish as sterile field.

5.1.4. Obtain vital signs and assess the level of consciousness. Attach on continuous cardiac monitoring.

5.1.5. Position the patient in Trendelenburg or as directed by the physician

5.1.6. Perform hand scrub using 4% chlorhexidine and wear sterile PPE.

5.1.7. Assigned staff will be a scrub nurse while other staff can be a circulating nurse with PPE during the procedure.

5.1.8. Pre flash all the central line device parts with heparinized normal saline under aseptic technique.

5.1.9. Assist in the performance of procedure as directed by the physician such as skin preparation, insertion, suturing and securing all lumens of catheter under strict sterile technique.

5.1.10. Connect all lumens to pre flashed three-way stopcock.

5.1.11. Apply op-site (transparent adhesive tape) dressing and label it with date and time.

5.1.12. If inserted in neck (Internal Jugular Vein), chest (Subclavian Vein or axillary vein), inform radiology department to perform chest x-ray as per physician before using the line.

5.1.13. The physician may confirm placement of the line by aspirating blood for Venous Blood Gases.

5.1.14. Position the patient as directed by the physician.

5.1.15. Dispose the equipment appropriately.

5.1.16. Perform hand hygiene. (Refer to APPH-IPC--0005 Guidelines in Hand Hygiene).

5.1.17. Assess vital signs and note any changes from the baseline.

5.1.18. Assess for any oozing, bleeding or hematoma over insertion site.

5.1.19. Document the date, time, site, condition, and patency of line, how many lumens, problems encountered with insertion if any and patient's tolerance during the procedure in Doctors Order Sheet Form, Nurses Progress Notes Form in the MCC and Infection Control Form.

5.2. **CVAD Care**

5.2.1. Physician writes the order.

5.2.2. Explain the procedure to the patient.

5.2.3. Gather all equipment needed.

5.2.4. Obtain vital signs and assess the level of consciousness. Attach on continues cardiac monitoring.

5.2.5. Place the patient in supine position.

5.2.6. Perform hand hygiene and don non-sterile gloves.

5.2.7. Open the dressing pack and add required equipment without any contamination.

5.2.8. Remove and discard previous dressing. Observe any signs of infection, inflammation or catheter leakage/damage

5.2.8.1. Notify the physician immediately If there's any sign of infection or catheter leakage/damage.

5.2.9. Perform hand hygiene and don PPE.

5.2.10. Clean exudates from insertion site with 2% Chlorhexidine/70% alcohol antiseptic solution in a circular motion from insertion site outwards approximately 10cm.

5.2.10.1. Antiseptic solution must be allowed to dry for maximum effect which takes 2 minutes approximately.

5.2.11. Apply op-site (transparent adhesive tape) dressing and label with date and time.

5.2.11.1. If site is oozing apply sterile gauze and transparent adhesive dressing.

5.2.12. Dispose the equipment appropriately.

5.2.13. Perform hand hygiene.

5.2.14. Document in Nurses Progress Notes Form in the MCC.

5.3. **CVAD Routine Flushing**

5.3.1. Explain the procedure to the patient.

5.3.2. Gather all equipment needed.

5.3.3. Perform hand hygiene and don non-sterile gloves.

5.3.4. Clamp any lumens not being accessed. Don't clamp lines with vasoactive infusions running.

5.3.5. Clean the access point with 2% ChlorhexidinelT0o/o alcohol antiseptic solution for at least 15 seconds and allow air to dry.

5.3.6. Release clamped lumen and flush the lines with l0 ml of 0.9% Sodium Chloride using

pulsatile action after each access or weekly if not in use.

5.3.6.1. If resistance is noted stop and re-clamp the lumen. Label the lumen "blocked" and inform the physician immediately.

5.3.7. For Heparin Locking the CVADs:

5.3.7.1. 5mL heparinized saline (50 Units/5ml) per lumen will be used.

5.3.8. Lock the port when not in use.

5.3.9. Dispose the equipment appropriately.

5.3.10. Perform hand hygiene.

5.3.11. Document in Nurses Progress Notes Form in the MCC.

5.4. **CVAD Removal**

5.4.1. Physician writes order for removal of CVAD

5.4.2. Explain the procedure to the patient.

5.4.3. Gather all equipment needed.

5.4.4. Place the patient in supine position.

5.4.5. Turn off all infusions and clamp all lumens.

5.4.6. Perform hand hygiene.

5.4.7. Open work field and add required equipment.

5.4.8. Perform hand hygiene and don PPE with non-sterile gloves.

5.4.9. Disconnect lines and carefully remove existing dressing while observing the site for any signs of infection. If infection is suspected, swab the site and send it for culture and sensitivity.

5.4.10. Change the existing gloves with sterile gloves.

5.4.11. Clean site with 2% Chlorhexidine/70% alcohol antiseptic solution in a circular motion from insertion site.

5.4.12. Cut and remove the sutures.

5.4.13. Ask the patient to perform Valsalva maneuver to minimize the risk of air embolism.

5.4.14. With the use of non-dominant hand hold a piece of gauze at the insertion site while dominant hand slowly removes the catheter.

5.4.15. Apply continuous and direct pressure for a minimum of 5 minutes. Then apply Opsite (Transparent Adhesive Tape)

5.4.16. Inspect catheter for intactness.

5.4.17. Dispose the equipment appropriately.

5.4.18. Perform hand hygiene.

5.4.19. Document the procedure and date of removal in Medica Cloud Care (MCC) system, Nurses

6. MEASUREMENT

6.1. 100% of the medical and nursing staff are well oriented and adheres to this policy.

7. REFERENCES /CROSS- REFERENCES

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8. ATTACHMENT

8.1. Central Venous Access Device Management Check Off

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design